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Growing fast and steady in space: Distributed rapid self-reconfiguration motion planning optimization methods for swarm intelligent space modular self-reconfigurable satellites

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ABSTRACT

The large-scale self-reconfiguration of Space Modular Self-Reconfigurable Satellites (SMSRS) faces critical challenges in the computational complexity of global assignment and motion space problems, low efficiency, as well as the low completion rate associated with the hollow structure, severely limiting their application in extraterrestrial infrastructure construction. In this study, we propose a Rapid Self-Reconfiguration Motion Planning Optimization (RSRMPO) method based on the distributed framework, specifically designed for pivoting cube modular systems. The proposed framework resolves the dual bottlenecks of near-optimal NP-complete problem computational demands and hollow structure formation through a decoupled optimization strategy: assignment optimization at the high-level module ensures structural integrity, while motion planning optimization at the execution layer guarantees swarm fast collision-free coordination. This study contains a Virtual Connection technique, a Degeneration-Growth Assignment (DGA) method, a real-time Feasible Space map (FS_map) generation technique, an energy-cost-optimized A* path planning algorithm, and a collision avoidance technique. In reconfiguration with size spanning 10-1,200 modules, RSRMPO framework demonstrates less planning time and energy consumption and achieves a higher completion rate compared to the Graph-Based Configuration Search (GBCS) algorithm. Simulation experiments confirm that RSRMPO framework provides near-optimal solutions to the NP-complete reconfiguration problem, effectively improving computational efficiency and structural robustness, which establishes significant implications for orbital intelligent large-scale assembly,on-orbit service, and deep-space exploration missions.

Introduction

As spacecraft missions become increasingly specialized and complex [1], development costs rise, launch cycles lengthen, and the demands of specific or urgent orbital tasks often remain unmet [2]. In this context, the emergence of the Space Modular Self-Reconfigurable Satellite (SMSRS) offers considerable promise in addressing these challenges [3]. The concept of “satellite modularity” was introduced by DARPA’s “Phoenix Program” with the HISat spacecraft in 2011 [4]. Other examples of modular intelligent systems include the M-Block robots developed by MIT [5] in Fig. 1(a), UBot robots [6], and Seremo robots [7]. SMSRS systems offer key advantages such as reusability, interchangeability, and robust data interfaces [8,9].

The modular satellite can self-reconfigure according to the mission requirements in Fig. 1(b). Some novel space robotic systems, like 3-RRU [10], can self-reconfigure or self-assemble [11] into deployable configurations for high-precision maneuvering [12,13], orbital capture [14] and in-orbit assembly of a large aperture space telescope [15]. Moreover, there are deformable spacecraft that can actively reconfigure their structure using operable joints [16], and multiagent robot manipulator with strong control robustness [17–19], where and space transportation challenges can be tackled through collaborative reconfiguration using multiple space modular substructures [20]. Due to their low cost and flexibility, the SMSRS have emerged as versatile platforms for various large-scale space missions, including on-orbit maintenance and debris removal [21,22]. This method not only enhances spacecraft development efficiency but also significantly reduces launch costs by forming target spacecraft structures in orbit. Meanwhile, design schemes of the

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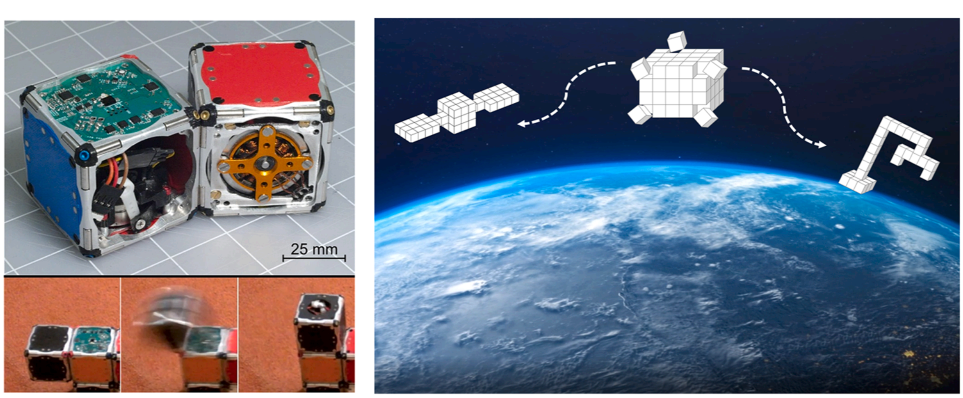
modular system have achieved considerable maturity in recent years, such as M-Block of MIT, which main contains a self-contained flywheel for the attitude control, power system, sensors for detection, communication module, docking mechanism (mechanical or magnetic), computers and other scientific payloads in Fig. 1(a).

The on-orbit self-reconfiguration (SR) of modular satellites has garnered great interest of researchers. Investigations into modular satellites have developed techniques for self-reconfiguration planning [23], which has been established as an NP-complete (Non-deterministic Polynomial) problem [24]. Thus, SR is divided into task allocation and path searching. In task allocation, Chirikjian et al. give the lower and upper bounds of the step of reconfiguration on the 2-D lattice [23], where the minimum steps depend on the Manhattan distance between the origin position and the final position, and the low bound of maximum steps is related to the current configuration geometry. Hou [25] and Rus [26] divide the SR planning task into multiple steps or completion levels for 2-D modules. It's flexible in choosing the intersection between initial configuration and target for maximum efficiency in reconfiguration. Fitch et al. [27] make the initial configuration melt into a kind of chainlike intermediate configuration, and the conversion process from the target configuration to the intermediate conformation is taken in reverse to complete the SR process. Daudelin et al. [28] offer a modular robotic system that can auto-execute high-level tasks via reactive reconfiguration, allowing it to adapt to unfamiliar settings. Li et al. [29] utilize Nash techniques to ascertain optimal judgments for formation systems. Guo et al. [30] studied the degrees of freedom of modular mechanisms utilizing the concepts of "Equivalent mechanism" and screw theory. The screw topological graph was used for the on-orbit assembly of the antenna mechanism. Dutta et al. [31] present a distributed solution that uses concepts from graph theory to address the merging of multiple individual or separated configurations into a final target configuration. Liu et al. [32] provide a framework for SMORES-type modular robots to efficiently self-assemble into a tree topology, where the target topology can be mapped into a planar pattern with optimal assignments based on the positions of the modules. Depending on the connected graph, a bipartite graph has been established between MSRSs, where an A* algorithm is used for MSRS configuration computation [33], and the method proves reconfiguration in O(n) moving steps within 50 modules. In path searching, Fitch et al. [34] realize obstacle avoidance within n modules in $O(n^2)$ time by constructing surface-free space and connecting tunnels in specific areas. Brandt et al. [35] evaluate two reconfiguration path planning algorithms, A* and RRT, for modular self-reconfigurable robots, ATRON. Liu et al. [36] present a method, homotopy-informed preprocessing (HIP), in the field of random sampling motion planning algorithms to resolve paradoxes by supplying supplementary information. Ye et al. [37] introduce a decentralized path planning method for modular satellites based on partially observable space, where the Q-learning method is used to train the module to reach the target position with less path cost. Obviously, the previous research has advantages for the reconfiguration with small-scale [25,38]. Moreover, if the configuration size increases and the structure becomes more complex, issues of structural integrity may arise, indicating that traditional centralized methods are inadequate. Sung et al. [39] present some unreconfigurable concave patterns. Song et al. [33] identify the hollow structure that obstructs subsequent module execution. The issue is inadequately addressed. Lindenmayer systems [40] are extensively utilized in mathematical modeling and computation [3,41], and the graph-theoretic framework has demonstrated superiority for the development of 2-D and 3-D structures [42], offering graphical interpretations that are also topological [43]. It inspires us to deal with the structural issue.

Furthermore, modular satellites are constrained by limited energy and computational resources owing to the microgravity conditions in space. The SR of MorphLine [38] enables modules to autonomously and rapidly transition between configurations in a distributed way [25], which may consume more energy primarily due to more transfer actions. A reconfiguration relationship diagram is used to find the shortest path [44], which shows how HSR changes between configurations, and also intensive computation. The heuristic search SR utilizes a heuristic function to guide the search for the reconfiguration path for each M-TRAN [25,45] module by maximizing the identity graph between configurations [46]. Bie et al. [47] provide a distributed control framework for UBot by presenting a method integrating cellular automata and L-systems for specific large-scale configurations, where the method shows a high reconfiguration completion rate. Because of limitations of path planning, there are more transfer steps involved. But the model's performance declines as the extent of alterations related to the training configuration increases. By predefining the joint path into a binary map in C-spaces, the self-collision avoidance strategy of the space modular self-reconfigurable satellite (SMSRS) can realize the offline path planning efficiently [48], which is based on the detection model. A hierarchical self-reconfiguration (HSR) method is proposed to improve the SR speed [49], which is derived from the plant cell generation process.

In this article, we are the first to focus on solving swarm space modular self-reconfigurable satellites' large-scale complex self-reconfiguration (SR) problems (over 1,000 modules) with energy optimization (fewer reconfiguration transfer steps) and improving the completion rate (preventing hollow structures). We propose an assignment optimization method combining a Virtual Connection Technique (VCT) and a Degeneration-Growth Assignment (DGA) algorithm, which can swiftly identify movable modules and prevent hollow structures by dynamically reassigning modules while maintaining global connectivity. A real-time motion planning optimization method is introduced that



(a) (b)

Fig. 1. The concept of space modular self-reconfigurable satellites. (a) The picture and flywheel-based pivoting motion of M-Block designed by MIT. (b) The on-orbit reconfiguration demonstration.

improves the overall efficiency of the reconfiguration optimal path search of swarm modules, integrating a Feasible Space Map (FS_map) for rapid collision domain identification, an energy-cost-optimized A* algorithm with parallelization, and a priority-based collision avoidance technique to prevent agent deadlock. Through random large-scale simulations, our methods are proven to have significant advantages of reducing the reconfiguration transfer steps and planning time and increasing the completion rate, covering configuration size from 10 to 1,200. The main contributions of this study are as follows: (1) A novel assignment algorithm for large-scale reconfiguration limited by the link condition, which is more efficient and faster than previous algorithms, is first proposed. This is different from prior works. The swarm assignment problem in large-scale reconfiguration is solved by two new methods: the virtual connection technique and the degeneration-growth assignment method. (2) A unique distributed motion planning algorithm for swarm modular satellites limited by the pivoting condition, which costs less calculation workload and fewer total transfer steps than traditional methods, is proposed. The swarm motion planning on the complex surface of satellites is divided into two independent problems: the real-time feasible motion space map and optimal transfer routes based on the enhanced A* algorithm. (3) The method provides the better solution to the large-scale reconfiguration of complex modular satellites with less total planning time, fewer total transfer steps, and a higher completion rate. Three typical application scenarios of the method are self-reconfiguration of the modular satellite with a camouflage satellite, a space telescope frame, and a great space manipulator arm.

The content of this article is organized as follows. Section 2 introduces the SMSRS and the problem of self-reconfiguration motion planning. Section 3 presents the assignment optimization algorithm utilizing the virtual connection technique and the degeneration-growth assignment method. Section 4 proposes the motion planning optimization algorithm involving a feasible motion space generation technique, an enhanced A* path planning algorithm, and a collision avoidance technique. In Section 5, random experiments are used to verify the feasibility and the superiority of our methods. Section 6 presents the conclusions of the study and the further application of the proposed methods.

Problem formation

The distributed strategy of SR can be divided into (1) high-level tasks: assignment planning and map generation; (2) low-level tasks: path planning, motion execution, and collision avoidance. This section explains the SMSRS and its motion conditions and the details of the SR problem. The hardware of the SMSRS includes additional payloads, such as interchangeable power modules, varied sensors, wireless modules, and a minicomputer, et al. Each SMSRS is an agent. And the grid is utilized to depict the module and positional data throughout the reconfiguration process. The grid is a cube identical in dimensions to the SMSRS. Table 1 summarizes the nomenclature of the main abbreviations used in this article. Table 2 summarizes the meaning of the main symbols used in this article.

Reconfiguration pivoting motion

The SR problem might be seen as a macroscopic outcome of the systematic movement of multiple modules on the surface of other modules, and the dynamics of a SMSRS is pivoting (rotating) about the edges they share with other modules, where the pivoting condition is determined by the neighbor modules and the observed space. In Fig. 2, there are two conditions to determine the motion type and the feasibility of the transfer between two positions, the support module and feasible space.

If the Manhattan distance between the two transfer positions, $P_a$toP_b, is eaqual to the length of one module, the transfer motion is pivoting by$\pi/2$, and there must be two support modulesCOF_s^candCOF_{s+1}^c, and a space restriction$R_{\pi/2}$.If the Manhattan distance between the two transfer positions is eaqual to twice the length of one module, the transfer motion is pivoting byπ, and there must be one support modulesCOF_s^c, and a space restriction$R_\pi$in Eqs. (1) and (2). $P_a^*$is the relative position in the coordinate of rotation axis L that is also the shared edge with its support module. $\Delta_L$is the position in the world coordinate. $\partial$is the angle of rotation, and $V_L^\partial$is the rotation matrix around shared axis L, where the rotation matrix is different for different rotation axes. The transfer motion is feasible only if the support module exists and the restriction matrix does not overlap the current configuration.

$$|P_a - P_b| = |x_a - x_b| + |y_a - y_b| + |z_a - z_b| \tag{1}$$

Table 1
Nomenclature.

Abbreviation	Long form
SR	Self-Reconfiguration
SMSRS	Space Modular Self-Reconfigurable Satellites
RSRMPO	Rapid Self-Reconfiguration Motion Planning Optimization
IMRP	Individual Module Reconfiguration Planning
FS_map	Feasible Space Map
GBCS	Graph-Based Configuration Search
VCT	Virtual connection technique
DGA	Degeneration-Growth Assignment
HDSR	Hormone-Distributed Self-Reconfiguration
HSSR	Heuristic Search Self-Reconfiguration
TBB	Tree-Based Branch and Bound
KM	Kuhn–Munkres algorithm
MSR	Modular Self-Reconfiguration
L-systems	Lindenmayer Systems

Table 2
Meaning of the main symbols.

Symbol	Meaning
$R_{\pi/2}$,R_π	Space restrictions of pivoting$\pi/2$andπ
V_i^k	Virtual active connection direction in thek^{th}step, the i is the module ID
υ	Virtual active motion index
B_i^k	Virtual passive connection state in the k^{th}step, the i is the module ID
ℓ	Virtual passive connection state parameter
P_i^k	The position of the module i in thek^{th}step
P_a	The position of the space
V_L^∂	The rotation matrix that rotates the angle ∂ about the L-axis
$\vec{w}$	Connection direction vector
A_s	Thes^{th}batch-ordered assignment
τ_s	The number of thes^{th}batch-ordered assignment
ϕ	A path searching solution
$\Theta_{i\to j}^\phi$	Intermediate position set from position i to position j in solutionϕ
$R(\phi,\eta)$	All transfer steps of the assignmentη
$\mathbb{R}$	Completion indication parameter
Υ	Observation state information
σ	Priority index of the observation position
$\vec{w}_a$	Observed position vector
$\vec{w}_a^{\otimes}$	Unobserved position vector
COF_i^c	Current configuration state of module i
COF_j^f	Final configuration state of vacany j
M	Global assignment set
Φ	Global path set
Λ	Overriding rule
$\mathbb{C}_k^f$	Growth configuration of final configuration in thek^{th}step
$\mathbb{C}_k^c$	Decline configuration of current configuration in thek^{th}step
Sr_i^c	Surface positions of current module$i$
p_e^c	Current Map position$e$
d	Euclidean distance of two current positions in the map
m_{s-1}^s	Motion cost from the parent position to the next candidate position
μ	Weight coefficient
U	Connected matrix of the map

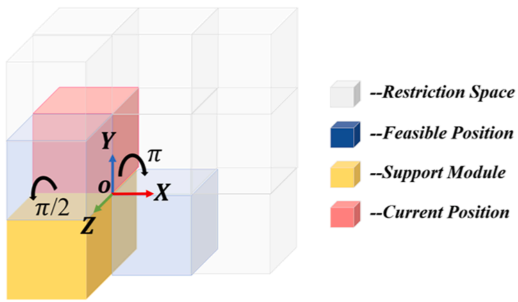


Fig. 2. Two basic reconfiguration pivoting motions and motion space.

$$\begin{cases} P_a = P_a^* + \Delta_L \\ P_b = V_L^{\partial} P_a^* + \Delta_L \\ P_{s+1} = P_s + (P_b - P_a) \end{cases}, V_{L_Z}^{\partial} = \begin{bmatrix} \cos\partial & -\sin\partial & 0 \\ \sin\partial & \cos\partial & 0 \\ 0 & 0 & 1 \end{bmatrix}, V_{L_Y}^{\partial} = \begin{bmatrix} \cos\partial & 0 & \sin\partial \\ 0 & 1 & 0 \\ -\sin\partial & 0 & \cos\partial \end{bmatrix}, V_{L_X}^{\partial} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\partial & -\sin\partial \\ 0 & 0 & \cos\partial \end{bmatrix} \tag{2}$$

$$R_{\pi/2} = \begin{Bmatrix} 2P_a - P_s \\ P_b + P_a - P_s \end{Bmatrix} \tag{3}$$

$$R_{\pi} = \begin{Bmatrix} 2P_a - P_s \\ 2P_b - P_s \\ P_a + P_b - P_s \\ 2P_a + P_b - 2P_s \\ P_a + 2P_b - 2P_s \end{Bmatrix} \tag{4}$$

$$FeasibilityConditions:$$

$$(1)\frac{\pi}{2} : P_s, P_{s+1} \in COF^c \& R_{\pi/2} \cap COF^c = \varnothing \tag{5}$$

$$(2)\pi : P_s \in COF^c \& R_{\pi} \cap COF^c = \varnothing$$

The current configuration$COF^c$is defined in Eqs. (6) (8) (9) and (10), including a detailed three-dimensional (3-D) positionP_i^c, completion indicator $\mathbb{R}_i$, virtual passive connection statusB_i^c, and virtual active connection $V_i^c$of the module$i$. The virtual active connectionV_i^cdenotes the motion condition, which contains the motion index$\upsilon$and the connection state matrix$W$. Motion indexυdetermines whether the module is movable, 1 for movable and 0 for unmovable, which is calculated in Algorithm 1. The typical unmovable state is shown in Fig. 3.

The completion indication parameter$\mathbb{R}_i$signifies whether the SMSRS $i$reaches its designated target configuration, with 1 for success and 0 otherwise. The virtual passive connectionB_i^cwill be described in detail in Sec 3.1. The final configuration, COF_i^f, defined in Eq. (7), contains positions extracted from commands and observation state informationΥ, which is mostly utilized for the prioritization of the local target position. $\vec{w}_a$and $\vec{w}_b$represent observed position vectors.$\vec{w}_a^{\otimes}$and $\vec{w}_b^{\otimes}$denote the unobserved position vectors. σserves as the priority index, with positive values indicating current reconfigurability, 0 denoting non-reconfigurability, and negative values signifying potential reconfigurability in Eqs (14) and (15).

Algorithm 1
Motion condition.

```
Input: Connection State Matrix W
Output: Motion index υ
    for all nonzero connection vector w_i^r in W
    if num({w_i^r})=2 & w_1^r + w_2^r = 0
    υ=0
    end if
    if num({w_i^r})≥4 & ∃(w_1^r + w_2^r + w_3^r + w_4^r = 0)
    υ=0
    end if
    The other remaining states mean moveable, υ=1
    end for
```

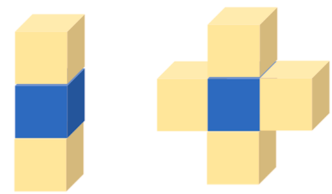


Fig. 3. Two typical states where the motion of the blue SMSRS is unfeasible. There is only one set of parallel connections, or there are two sets of parallel connections.

$$COF^c = \left\{ \left[Id_i, P_i^c, V_i^c, B_i^c, \mathbb{R}_i \right] \middle| i = 1, \ldots, n \right\} \tag{6}$$

$$COF^f = \left\{ P_j^f, \Upsilon_j^f \middle| j = 1, \ldots, n \right\} \tag{7}$$

$$P_w = [x_w, y_w, z_w]^T | w = 1, \ldots, n \tag{8}$$

$$\mathbb{R}_i = \begin{cases} 0, \text{unoccupied} \\ 1, \text{occupied} \end{cases} \tag{9}$$

$$\begin{matrix} V^k = \{\upsilon, W\} \\ B^k = \{\ell, W\} \end{matrix} \tag{10}$$

$$W = \left[\vec{w}_1^{x+}, \vec{w}_2^{x-}, \vec{w}_3^{y+}, \vec{w}_4^{y-}, \vec{w}_5^{z+}, \vec{w}_6^{z-} \right] \tag{11}$$

$$\| \vec{w} \| = \begin{cases} 1, \text{there is a virtual passive connection} \\ 0, \text{there is no virtual passive connection} \end{cases} \tag{12}$$

$$\ell = \begin{cases} 0, \sum_{r=1}^{6} \| \vec{w}^r \| = 0 \\ 1, \text{otherwise} \end{cases} \tag{13}$$

$$\Upsilon_j^f = \{\sigma, W\} \tag{14}$$

$$\sigma = \begin{cases} 0, \sum_{r=1}^{6} \| \vec{w}^r \| = 0 \text{or} \sum_{r=1}^{6} \| \vec{w}^r \| \geq 5 \text{or} \left\{ \sum_{r=1}^{6} \| \vec{w}^r \| = 4, \vec{w}_a^{\otimes} \| \vec{w}_b^{\otimes} \right\} \\ -1, \sum_{r=1}^{6} \| \vec{w}^r \| = 2, \vec{w}_a \| \vec{w}_b \\ 1, \sum_{r=1}^{6} \| \vec{w}^r \| = 1 \\ 2, \sum_{r=1}^{6} \| \vec{w}^r \| = 2, \vec{w}_a \perp \vec{w}_b \\ 3, \sum_{r=1}^{6} \| \vec{w}^r \| = 3 \\ 4, \sum_{r=1}^{6} \| \vec{w}^r \| = 4, \vec{w}_a^{\otimes} \perp \vec{w}_b^{\otimes} \end{cases} \tag{15}$$

The swarm SR optimization problem

In existing centralized strategy, the optimal SR motion planning problem involves selecting an appropriate *Id* assignment and sequence for positions in the final configuration and finding the optimal route for each assignment to minimize the objective function in Eq. (17). The computational complexity of matching all potential *Id* assignment and final configurations is$O(n!)$.When all assignments achieve global optimality, the minimum average theoretical SR steps can be computed for random reconfigurations of varying sizes and overlap rates, as per the KM algorithm illustrated in Fig. 4. The configuration size ranges from 10 to 1,000 modules, with each module comprising 10 pairs of random configurations, and the overlap rate spans from 10 % to 90 %. The 3-D spatial attributes of configuration-concavity and convexity, along with the distance between assignment modules and target positions, also affect the overall reconfiguration, and the average time step may not be appropriate to evaluate the performance of all algorithms under 3-D random reconfiguration. In face of random target configuration, the SR motion planning strategies primarily utilize graph matching and path optimization techniques to refine the *Id* assignment relative to the final configuration positions to reduce the total reconfiguration steps.

In our distributed strategy, there is a high level control which is responsible for the acquisition of status information of partial modules in current configuration, reconfiguration planning and decision-making, and the release of motion commands to the modules that need to move. Due to the movement and spatial constraints of SMSRS, the overall SR optimization process is subdivided into stages where each SMSRS moves along the surfaces of other modules, resulting in dynamic geometric structure changes in configuration and reconfiguration space. Consequently, dynamic assignmnent and path palnning are required based on the real-time configuration.

We study the objective function in Eq. (16) to minimize the total cost of the overall SR process, focusing on reducing the number of transfer actions. The solution depends on two key factors: the global assignment set$M$and corresponding path setΦin Eqs. (18) and (20). When the size of configuration$n$is small, we can enumerate all sequences$T(n)$, compute the number of reconfiguration steps associated with each sequence, and identify the reconfiguration allocation sequence with the minimal transfer steps. However, when the configuration size is excessively large, the reconfiguration of each sequence using enumerated types will incur significant computing costs. Nevertheless, the distributed strategy alone organizes the sequence of the local partial reconfigurable modules, and the computational demand for each planning is little. The A_s represents thes^{th}batch-ordered assignment, which contains$\tau_s$pairs of sequences, matching the local partial current configuration and target configuration in Eq. (19). The overall assignments affect the evolution of the reconfiguration space, influencing the reconfiguration route for each

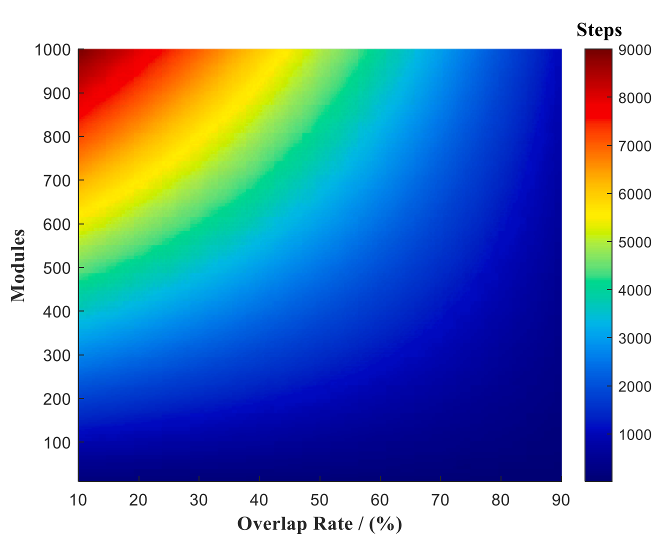


Fig. 4. The overlap rate, size of configuration, and space characteristics influence SR steps.

matching scenario under each assignment in Eq. (17). The$R(\phi, \eta)$, consisting of the current position $P^c$of module i,the final position $P^f$of the vacancy$j$, and the intermediate position set Θ^{ϕ} from P^ctoP^fas described in Eqs. (19) and (20), represents the path of theη^{th}assignment inA_s, where theϕdenotes a possible path-searching solution. In the istributed method, we conduct local sequence and path optimization for each local reconfiguration, enhancing local reconfiguration performance and aiming for global optimization through cumulative effects in Eq. (21), resulting in fewer reconfiguration steps and reduced planning time.

$$\{M^*, \Phi^*\} = \arg \min_{R(\phi,\eta)\in\Phi} \sum_{As} \sum_{\tau s} R(\phi, \eta) \tag{16}$$

$$M \in T(n) \tag{17}$$

$$M = \{A_s | s = 1, 2, \ldots\}, n = \sum_{s=1} \tau_s \tag{18}$$

$$A_s = \left\{ \left(P_i^c, P_j^f \right)_\eta \middle| \eta \in 1, \ldots, \tau_s \right\} \tag{19}$$

$$R(\phi, \eta) = \left[P_i^c, \Theta_{i\to j}^{\phi}, P_j^f \right] \tag{20}$$

$$\{A_s{}^*, \Phi_s^*\} = \arg \min_{R(\phi,\eta)\in\Phi_s} \sum_{\eta=1}^{\tau_s} R(\phi, \eta) \tag{21}$$

Assignment optimization algorithm

Virtual connection technique

The existing reconfiguration planning must ascertain the connection state of the configuration through the centralized graph connection calculation prior to reconfiguration, hence augmenting the computational demands in large-scale reconfiguration, which is obviously not suitable for large-scale configurational calculations. To enhance reconfiguration planning efficiency, we propose a virtual connection technique based on magnetic link sensors and the distributed framework to facilitate rapid identification of connection states among modules and minimize the repetitive calculations of the connection status of the reconfiguration intermediate process in Fig. 5(a).

Before reconfiguration, the SMSRS system has its own body coordinate (origin position), and each module has its relative space position. Once the system receives the external target reconfiguration information, which contains position data based on the body coordinate of the SMSRS, each module assesses its relative space position in the current configuration against the target configuration. For modules that intersect with the target configuration, a virtual connection request signal will be generated, which is transmitted to truly connected modules and recorded by these modules, and then transmitted to the bottom-layer modules in the other connection directions of the upper-layer modules until the signal is transmitted to the outermost modules of the configuration, known as the root modules. Each transmission of a signal will be annotated with the count of the signal layers. If the index of signal layers between the two modules is identical, the signal does not establish a virtual connection in this direction. If the signal layers of the two modules are contiguous, the virtual active connection is established in the bottom-layer module, where the direction is from the bottom layer to the upper layer, and the virtual passive connection status is established in the upper-layer module.

The virtual passive connection status comprises a status index and a status matrix as delineated in Eq. (10). The status matrix comprises six connection direction vectors, namely $x+$, x-, $y+$, y-, $z+$, and z- in Eq. (11). If a virtual passive connection exists, the state index is set to 1 in Eq. (12). The state matrix will be employed to assess the virtual connection status of the module in Eq. (13).

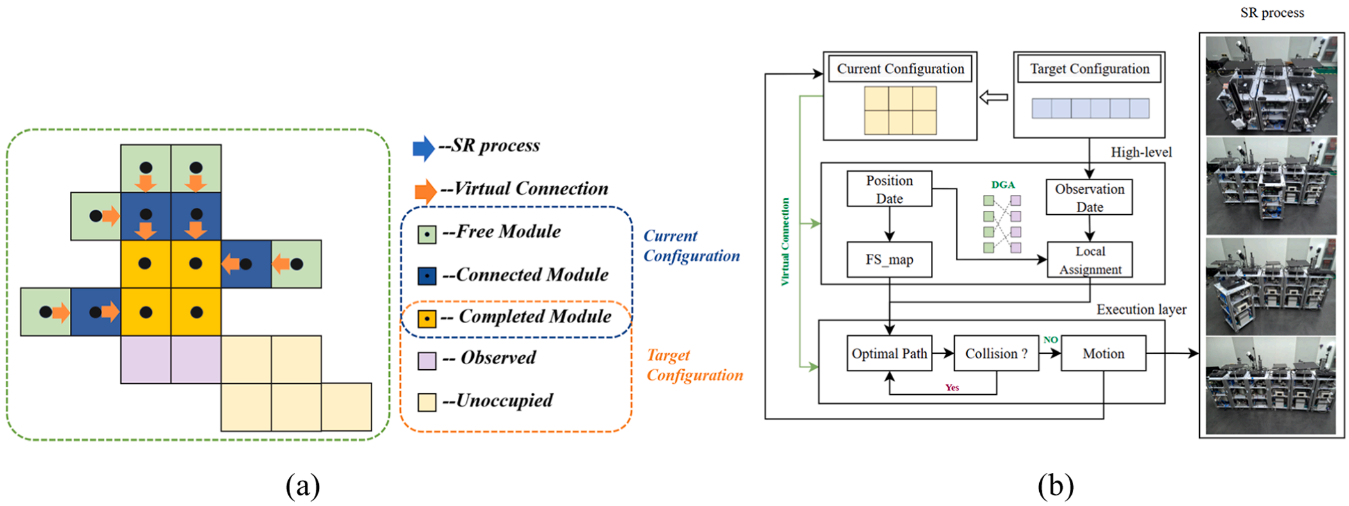


Fig. 5. The self-reconfiguration planning framework. (a) The virtual connection. (b) The modules can convey status information, position data, and observational details. The high-level module analyzes the data and generates the real-time task. The execution module obtains the target instruction and mapping data for SR. The observed position is the growth coniguration.

Dstributed control framework

We develop a distributed optimization method framework, where the transmission of the status data is consistent with the direction of the connection, and each module contains the position and connection data of all modules behind it. Upon receiving a command containing solely positional data regarding the target configuration, all modules compare their current positions with those in the target configuration. The SMSRS in the current configuration has three possible states:

(1) Free state;
(2) Immovable state;
(3) Completed state.

The positions in the final configuration can be in either an unoccupied or occupied state. Under the planning of the high-level module, free SMSRSs (the execution layer module) will be allocated to the unoccupied segments of the final configuration, as monitored by modules on the surface of the current configuration. The high-level module will compute the current feasible space map utilizing the interconnection data of all modules and supplies it to the execution layer module. The execution layer module will utilize the map to identify the optimal path, perform parallel motion, and prevent collisions. This process is repeated in a loop until the final configuration is complete, as illustrated in Fig. 5 (b).

Degeneration-growth assignment method

In large-scale SR processes, the assignment matching planning is a primary challenge. Local behaviors of the configurations are assigned to address this, enabling the entire system to converge toward the desired macro goal. Traditional approaches often focus on graph topology or distance matching by centralized methods; however, graph topology can result in high computational costs, while distance matching may lead to suboptimal local solutions. To overcome these limitations and achieve a significant gradient, the Degeneration-Growth Assignment (DGA) method, an improved fractal method, is introduced into SMSRS in Fig. 6. This method leverages the character-overriding mechanism of the enhanced L-systems to topologically describe the SR process [40,47] in Eq. (22).

The*Axiom* : $\mathbb{C}\Lambda$ represents the overriding mechanism of the SR process, where$\mathbb{C}^o$represents the original configuration state, $\mathbb{C}^c$represents

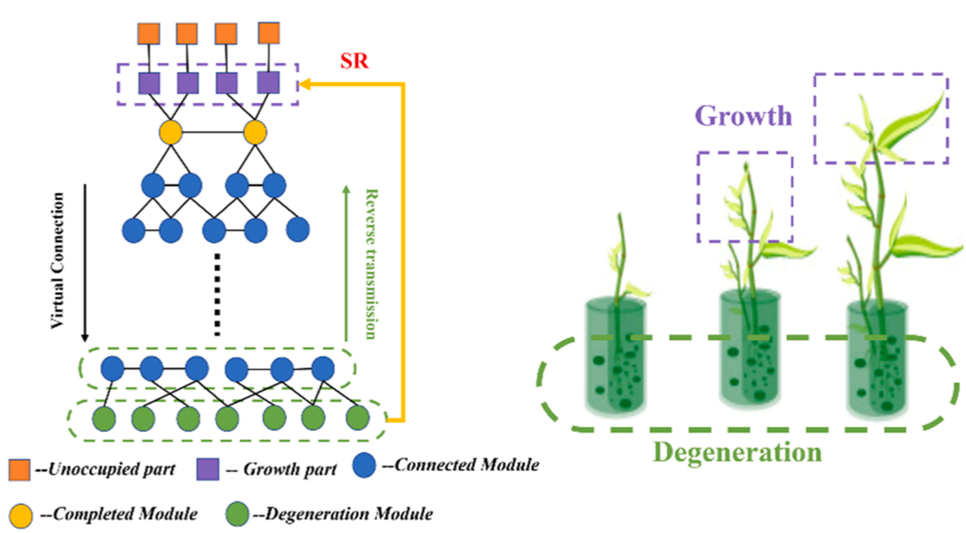


Fig. 6. The fractal SR process of the enhanced L-systems, like plant growth, takes in elements from the roots and gradually builds on existing structures.

the degeneration configuration of the current state.$\mathbb{C}^f$,which is get by limited observation of the SMSRS, represents the growing configuration of the final state andΛdenotes the SR rule. The rule contains two overrides, $+\mathbb{C}_k^f$and $-\mathbb{C}_k^c$.

The $-\mathbb{C}_k^c$represents removing degeneration configurations. In general,$-\mathbb{C}_k^c$ contains modules whose $\mathbb{R}=0$,$B^k,\ell=0$ and $V^k,\upsilon=1$ in the k^{th} generation. In certain complex instances$\left(\mathbb{R}=0,B^k,\ell=0, and V^k,\upsilon=0\right)$, outside modules fail to satisfy the motion condition, changing the execution layer through reverse transmission to reidentify the movable module. Subsequently, upon the completion of this reconfiguration cycle, the virtual connection among the overall configurations is rebuilt. Conversely, $+\mathbb{C}_k^f$ generates the growth coniguration, $\Upsilon_j^f:\sigma>0$,akin to cellular growth, creating an overlapping configuration that includes the geometric envelope of both the current and final configursations. This process attempts to fill any gap in $\mathbb{C}_k^f$, even if the numbers are not equal, as illustrated in Fig. 6. The overriding mechanism continues iteratively until the final configuration is fully achieved. The assignment matching in the SR process is the configuration assignment between$\mathbb{C}_k^f$ and $\mathbb{C}_k^c$. Therefore, the SR process can be broadly categorized as generating and eliminating various local configurations through the improved L-system method.

Outer modules that sense the local target configuration$\mathbb{C}_k^f$pass the information to free modules in$\mathbb{C}^c$. After obtaining the local generation and elimination configurations, the set of SMSRS, $\mathbb{C}_k^c$, and final positions, $\mathbb{C}_k^f$, are matched using the optimal matching algorithm developed by KM [50,51]. The minimum SR step for the movement of a module in a 2-D discrete space is determined by the Manhattan distance between its current and final positions [23]. This study extends the assignment-matching method to a 3-D space using the minimum Manhattan distance and KM algorithm. In Eq. (23), the Manhattan distance$Dw(i,j)$, represents the spatial distance between each free SMSRSP_i^c, whose$\mathbb{R}_i$ is 0, and each target final position P_j^f. To facilitate the computation in the KM algorithm, the match gain of $Dw(i,j)$ are negated.

$$
\begin{aligned}
&Axiom:\mathbb{C}\Lambda\\
&Rule:\Lambda\rightarrow\left[+\mathbb{C}_k^f-\mathbb{C}_k^c\right]\Lambda\\
&Gen_0:\mathbb{C}^o\Lambda\\
&Gen_1:\mathbb{C}^o\left[+\mathbb{C}_1^f-\mathbb{C}_1^c\right]\Lambda\\
&Gen_2:\mathbb{C}^o\left[+\mathbb{C}_1^f-\mathbb{C}_1^c\right]\left[+\mathbb{C}_2^f-\mathbb{C}_2^c\right]\Lambda\\
&\cdots\\
&Gen_k:\mathbb{C}^o\left[+\mathbb{C}_1^f-\mathbb{C}_1^c\right]\ldots\left[+\mathbb{C}_k^f-\mathbb{C}_k^c\right]\Lambda
\end{aligned}
\tag{22}
$$

$$
Dw(i,j)=-\left|x_i^c-x_j^f\right|-\left|y_i^c-y_j^f\right|-\left|z_i^c-z_j^f\right|
\tag{23}
$$

$$
boudary:l(c_i)+l\left(f_j\right)=Dw(i,j)
\tag{24}
$$

$$
\text{initial}:\begin{cases}l(c_i)=\max\limits_{f\in\mathbb{C}_k^f}Dw(c_i,f)c_i\in\mathbb{C}_k^c\\ l\left(f_j\right)=0f_j\in\mathbb{C}_k^f\end{cases}
\tag{25}
$$

$$
\delta_l=\min_{c_i\in S,f_j\in T}\{l(c_i)+l(f_i)-Dw(i,j)\}
\tag{26}
$$

$$
l=\begin{cases}l(c_i)-\delta_l c_i\in\mathbb{C}_k^c,c_i\notin S\\ l\left(f_j\right)+\delta_l f_j\in\mathbb{C}_k^f,f_i\notin T\end{cases}
\tag{27}
$$

$$
A_s^t=\left\{\left[\mathbb{C}_k^{c*},\mathbb{C}_k^{f*}\right]\right\}
\tag{28}
$$

In the optimal matching algorithm, two sets are defined: the set Γ, representing the SMSRS in $\mathbb{C}_k^c$, eligible for matching, and theΨset, representing the regions, $\mathbb{C}_k^f$, need to be occupied. The $l(c_i)$ denotes the label of SMSRS $i$in set Γ, and $l\left(f_j\right)$ denotes the label of the unoccupied region $j$in set Ψ. The boundary and initial conditions for determining the optimal matching are specified in Eqs. (24) and (25). Next, the Hungarian method is applied, incorporating these boundary conditions, to establish the complete matching between$\mathbb{C}_k^c$and$\mathbb{C}_k^f$, which has the maximum sum of weights. $\mathbb{C}_k^{c*}$and$\mathbb{C}_k^{f*}$, which correspond in equal quantities, is constituted. If there are no corresponding repetitions, then$\left(P_i^c,\ P_j^f\right)$represents the current optimal match. Otherwise, the label$l$is modified. Let $S:S\subset\Gamma$and $T:T\subset\Psi$, denote subsets of $\mathbb{C}_k^c$ and $\mathbb{C}_k^f$ that were uninvolved in the initial matching process according to Eqs. (26) and (27). When the current matching computation is completed, meanning the maximum match gainDw, we will prioritise the target sequence based on the magnitude of $\Upsilon_j^f:\sigma$to prevent the ignoring of concave structures and the emergence of hollow formations resulting from random and chaotic distribution. Ultimately, a current assignment$A_s^t$is formed, which contains matching information with target priority sequence. The high-level module passes this command to the execution layer module, the execution layer module plans the motion of the assigned target position for local configuration.

Motion planning optimization algorithm

Real-time feasible space map generation technique

A typical volumetric description for reconfigurable satellites is an occupancy grid space, which discretizes the position occupied by the SMSRS into a limited number of cells. This space is a voxel representation of the satellite's configuration. The grid cell's size is fixed at the SMSRS's size for matters with SR. Due to the limited observation range of the SMSRS, it can only detect the direction and position information of nearby modules and limited observed space using the sensors. In the zero-gravity space environment, the SMSRS can only move on the surface of the configuration.

To enhance the pathfinding for batch modules' reconfiguration, we offer a technique for generating finite feasible space maps based on connectable surface positions of immovable modules in the current configuration by communication. The connectable surface is generated by existing free surfaces$Sr^c$of each module in six directions, as illustrated in Fig. 7(a). Thep^crepresents the center point of the connectable surface, the map point. The$P_s$denotes the position of the SMSRS connection corresponding to the map pointp_s^c. Each position$p^c$of each SMSRS' six directions has the potential to be connected (active or passive connection) in Eqs. (29) (30). If a position inSr_i^cconflicts with the$Sr^c$of other SMSRS, it is removed from the setp^c on both sides, thereby forming the current feasible space map (FS_map^c) until all conflicts are resolved, as shown in Fig. 7(b). The FS_map^{c}represents the current feasible space that the SMSRS can navigate, which includes candidate map point p_e^c, identity Id_e^c, and its parent point $P^c(P^c\in COF^c)$, as detailed in Eq. (31).

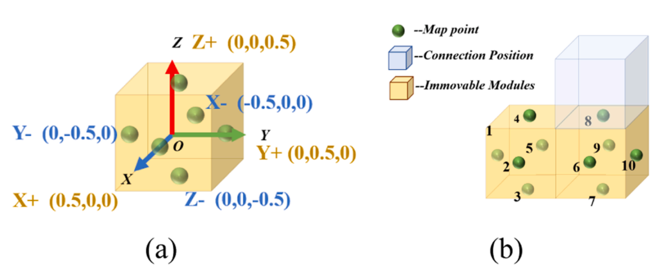


Fig. 7. FS_map. (a) Surface extensions in six directions for the SMSRS. The green points indicate potential connection positions for other SMSRS. (b) FS_map of two SMSRS.

The FS_map^c is dynamic and evolves as the configuration changes during the SR process.

$$Sr_i^c = \{p^{x+}; p^{x-}; p^{y+}; p^{y-}; p^{z+}; p^{z-}\} \tag{29}$$

$$\begin{aligned} p^{x+} &= P_i^c + [0.5, 0, 0]; p^{x-} = P_i^c + [-0.5, 0, 0]; \\ p^{y+} &= P_i^c + [0.5, 0, 0]; p^{y-} = P_i^c + [0, -0.5, 0]; \\ p^{z+} &= P_i^c + [0, 0, 0.5]; p^{z-} = P_i^c + [0, 0, -0.5]. \end{aligned} \tag{30}$$

$$\text{FS\_map}^c = \left\{ \left[Id_e^c, p_e^c, P_i^c \right] | e = 1, 2, \ldots \right\} \tag{31}$$

$$d(p_u^c \leftrightarrow p_v^c) = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2 + (z_u - z_v)^2} \tag{32}$$

To facilitate path searching, a connected graph of the FS_map is established based on the SMSRS dynamics, where the connection states of the points are clearly defined on the map. The connection relationships in the graph are represented by a matrix U, where the l denotes the side length of the SMSRS, as outlined in Algorithm 2. Each point p_e^c in the FS_map^c is individually calculated to determine if it can be reached under the movement conditions of the SMSRS, as shown in Figs. 6(a) and (b). Euclidean distance $d(p_u^c \leftrightarrow p_v^c)$ and $d(P_u^c \leftrightarrow P_v^c)$ is used to evaluate the connection relationship between points p_u^c and p_v^c, as described in Eq. (32), where P_u^c and P_v^c are parent points of p_u^c and p_v^c. If the distance is within the range limited by the transfer motion of SMSRS, p_u^c and p_v^c are considered connected. Different types of transfer motions are marked with unique identifiers. Two points p_u^c and p_v^c on a map are equivalent if they correspond to the same connection position P_s. Consequently, the SMSRS can navigate the FS_map^c based on the matrix U.

Path planning algorithm

The A* search algorithm is particularly advantageous for problems that require path searching in large or finite search spaces, and it has low computing resource requirements, suitable for embedded devices. If the problem has a solution, it will be able to find the optimal solution. In reconfiguration path problem, the A* search algorithm may effectively use its benefits when integrated with the real-time feasible map. In this study, shortening the SR path and fewer steps translate to lower energy consumption. To achieve the goal of minimizing transfer steps, we combine an enhanced A* algorithm with motion space generation technique to improve the optimal path finding of batch SMSRSs. In Algorithm 3, all the connection surfaces of the free module and their corresponding center points are confirmed, and the candidate points of each center point are obtained according to the matrix U, which are the initial candidate map points p_s^c, is used in the *Open* set of the A* algorithm. If $U(p_s, p_{s-1}) = 0$, it means that the two map points have the same connection position P_s. Then it will take P_s as the center point and re-expand reachable points to *Open* set.

Algorithm 2
Connected matrix U.

Input: Current feasible space map.
Output: Connected matrix U.

for all p^c such that ($p^c \in$ FS_map^c) **do**
if $d(p_u^c \leftrightarrow p_v^c) = l$
if $d(P_u^c \leftrightarrow P_v^c) = l$
$U(u, v) = \pi/2$
end if
if $d(P_u^c \leftrightarrow P_v^c) = 2l$
$U(u, v) = 0$
end if
end if
if $d(p_u^c \leftrightarrow p_v^c) = \sqrt{l}/2$
if $d(P_u^c \leftrightarrow P_v^c) = 0$
$U(u, v) = \pi$
end if
if $d(P_u^c \leftrightarrow P_v^c) = \sqrt{2}l$
$U(u, v) = 0$
end if
end if
if $d(p_u^c \leftrightarrow p_v^c) > l$ or $d(p_u^c \leftrightarrow p_v^c) = 0$
$U(u, v) = \inf$
end if
end for

Algorithm 3
Improved A* path planning algorithm.

Input: $\left(P_i^c, P_j^f\right)_\eta$, COF^c, U and FS_map^c
Output: Route $R(\phi, \eta)$

Expand **all** reachable points p_s^c of P_s into *Open*
while *Open* $\neq \emptyset$ and $P_s^\phi \neq P_j^f$
for all p_s^c in the *Open* **do**
Compare P_{s-1}^ϕ to P_s^ϕ
if $P_{s-1}^\phi = P_s^\phi \leftarrow U(s-1, s) = 0$
Expand **all** reachable points $p_s^{c'}$ of P_s^ϕ into *Open*
Else
Evaluate p_s^c: $F(p_s^c) = G(p_s^c) + H(p_s^c)$
end if
if $P_s^\phi = P_j^f$
Return *Closed* and form route
end if
if p_s^c cost the least
Closed $\leftarrow p_s^c$
$G(p_s^c) \leftarrow G(p_{s-1}^c) + G(p_{s-1}^c \leftarrow p_s^c)$
end if
Open $\leftarrow$ Expand points p_{s+1}^c of p_s^c by U
end while

$$F(p_s^c) = G(p_s^c) + H(p_s^c) \tag{33}$$

$$H(p_s^c) = d\left(p_s^c \leftrightarrow P_j^f\right) + \mu \cdot m_{s-1}^s \tag{34}$$

A cost function $G(p_s^c)$ is used to measure the path cost from the candidate p_s^c in the connected matrix U and FS_map^c, to the parent position p_{s-1}^c, calculated using the Euclidean distance. A transfer gradient heuristic function $H(p_s^c)$ includes the Euclidean distance to the final position, and the the motion cost m_{s-1}^s, which is the number of transfer steps from the parent position p_{s-1}^c to the candidate p_s^c, and μ is the weight coefficient in Eq. (34). In this article, μ is often 0.707. $F(p_s^c)$ denotes the length of the route in Eq. (33). When candidate routes have identical values of $F(p_s^c)$ and $G(p_s^c)$, the route with lower $H(p_s^c)$ value. Points awaiting expansion are stored in the *Open* set, sorted by their $F(p_s^c)$ values. The point with the minimum $F(p_s^c)$ value is moved to the *Closed* set if it is a candidate point and not already present in the *Closed* set. The route, denoted by p^c, is then converted to the position $P_s^\phi \in \Theta_{i\to j}^\phi$ of the SMSRS in the configuration space, as shown in Algorithm 3. At the same time, the derived restriction space corresponding to the route is also formed, which is used for obstacle avoidance judgment. If there are multiple free modules in the same batch of local reconfiguration tasks, the optimal transfer path of these modules can be calculated in parallel.

Collision avoidance technique

Upon completed path searching in each free module, owing to the SR process of multiple SMSRS within the same batch, there may be two types of collision conflicts:

1) *Endpoint Conflicts*, primarily occurring at the starting or target points of each module's route and interfering with the routes of other modules;
2) *Space Conflicts*, which occur at the route where the derived restriction space of one module interferes with the route of another module.

If no collision exists, they can move in parallel. In order to avoid possible collision, we present a movement priority criterion based on the observable space and the route information. The priority is calculated between two modules once an SMSRS observes another module, which is in its next location or in its corresponding derived restriction space. following these rules below:

Criterion 1: If module A's starting point conflicts with module B's next location, module A gets higher priority.

Criterion 2: If module A's target point conflicts with module B's next location, module B gets higher priority.

Criterion 3: If the target point of a finished module A is on the unfinished module B's next location, module A inherits module B's rest of the route and gets higher priority.

Criterion 4: If the location of an unfinished module A is in the derived restriction space of another unfinished module B, module A gets higher priority and continues moving until it leaves the current derived restriction space of module B.

Criterion 5: If the location of a finished module A falls within the derived restriction space of an unfinished module B, module B designates the subsequent location as infeasible, updates itsFS_mapand connection matrix$U$, and reconfigures its route using Algorithm 3.

To evaluate the performance of algorithms in renconfiguration completion, we introduce the completion rate in Eq. (35), where theεdenotes the completion rate, the$N$is the size of the target configuration,$n_{overlap}$represents the the size of the overlap configuration between the target configuration and the original configuration, and$n_{finished}$represents the quantity of modules that finished the reconfiguration.

$$\varepsilon = \frac{n_{finished}}{N - n_{overlap}} \times 100\% \tag{35}$$

Experiments

For the reconfiguration of the SMSRS, we focused on the performance of two reconfiguration methods on assignment and path planning across different scale reconfigurations in this section. Reconfiguration planning time, number of the transfer step, and completion rate are our primary concerns, which represent the mobility, the energy consumption, and the expected state to SMSRS in space.

Experiment setting

To assess the efficacy of the SR method presented in this article, we executed three groups of experiments.

(1) The first group aims to evaluate the efficiency of the two methods (the graph-link method and VCT) that generate movable modules. The experimental configuration size varies from 10 to 1,000 modules, with 10 random configurations at each size, and the two methods independently compute the movable modules in the current configuration. (2) The second group is used to evaluate the performance of the two methods in reconfiguration path planning. Each experiment comprises three sets of the same assignments that have been predetermined. The configuration size ranges from 10 to 1,200 modules, with 10 sets of random configurations for each size. (3) The third group features configuration sizes ranging from 10 to 1,080 modules, with each size comprising 10 pairs of random configurations with 50 % overlap. This setup is designed to simulate the self-reconfiguration process between two random unknown configurations with the same size and assess the overall effectiveness of three SR methods in assignment and motion planning. In where partial configurations comprise random concave vacancies as shown in Fig. 10(c). This is specially designed to evaluate the assignment performance of three SR methods in the structural integrity. The trials evaluated three SR methods: 1) The graph-based configuration search (GBCS) method [33], which is a centralized algorithm. 2) The individual module reconfiguration planning (IMRP) method based on our frame as a control group. 3) The rapid self-reconfiguration motion planning optimization (RSRMPO) method (our method) for swarm modules. The IMRP method is a control group to GBCS method and RSRMPO framework in reconfiguration process. Three performance metrics—the whole planning time, transfer steps, and completion rate—were evaluated.

Experiment results

Fig. 8 shows the results of the two methods in generating movable modules. When the size of the configuration is less than 150 modules, the average time cost of generating movable modules by the two algorithms is almost equivalent. As the time complexity of the graph-link method is $O(n^2)$ by DFS, the calculation speed is as fast as VCT in small scale, which is mainly through the parallel computing module connectivity. Furthermore, when the configuration size increases, the VCT is faster than the graph-link method, resulting in a significant time benefit. Under the configuration of 1000 modules, our method reduces the computation time by 51.2 % compared to the graph-link method.

For reconfiguration path planning in the same assignments, Fig. 9(a) illustrates the total average number of transfer steps calculated by the CS and MPO methods. It can be seen that under the scale of 100 modules, the reconfiguration planning effects of the two methods are similar; however, as the configuration size increases in complexity, the MPO method demonstrates superior capability in planning reconfiguration path compared to the CS method. This advantage primarily arises from our heuristic function, which incorporates the motion cost, thereby guiding each module to seek paths with minimal transfer steps. Meanwhile, multiple modules operate in unison, thereby decreasing the alteration of configuration features resulting from constant reconfiguration. Consequently, the cost of searching the configuration space is diminished, leading to lower energy consumption during reconfiguration. Regarding the average planning time of a single module, the CS method does not initiate planning for the subsequent module until the reconfiguration planning for the preceding module is complete. Conversely, the MPO method enables each designated module to concurrently strategize during the ongoing reconfiguration cycle, resulting in considerable time efficiency. We extracted the average planning time curve of a single module in the CS method, which reveals that the MPO algorithm decreases the reconfiguration planning time for each module by an average of 67.2 % when configured with 1,200 modules, as shown in Fig. 9(b). FS_map makes SMSRS search the reconfiguration path in the limited space, which reduces the search range to a certain extent.

In 50 % overlap rate, we simulated the SR process between two

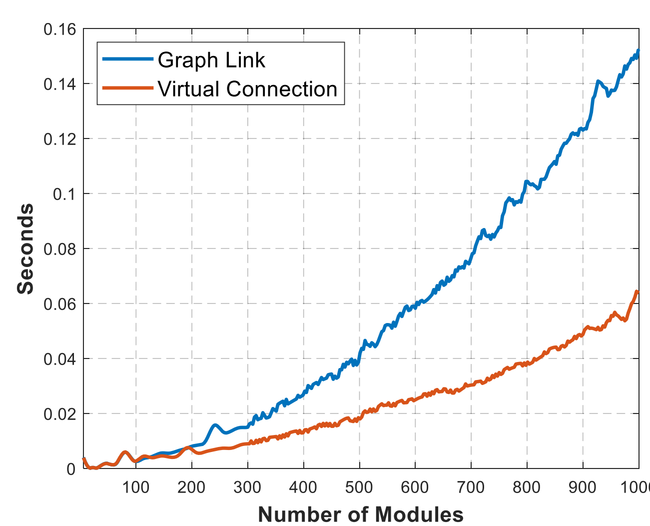


Fig. 8. The average time of generating movable modules.

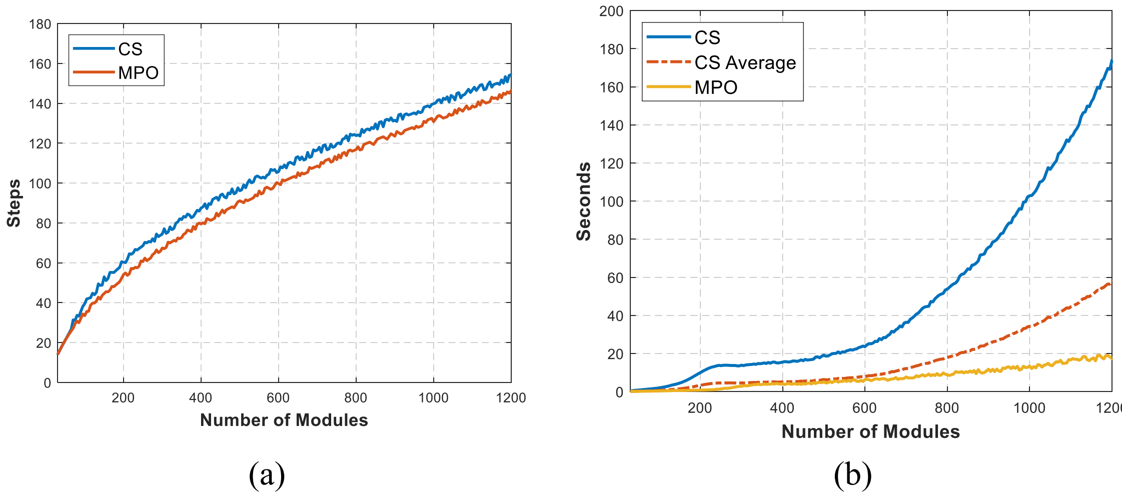


Fig. 9. The reconfiguration path planning for three sets of predetermined assignments. (a) The average step of the reconfiguration process. (b) The average time of generating movable modules. CS represents the configuration searching method. CS Average represents the average time per module reconfiguration plan. MPO is our motion planning optimization method.

random unknown configurations with the same size ranging from 10 to 1080 modules. The average value of the SR experiment results for each set of successful trials is shown in Figs. 10(a) and 10(b). When the configuration size is 50 modules, the assignments produced by GBCS and DGA differ little, and the number of transfer steps under the three methods is almost the same. However, IMRP decreases planning time by 50 % and RSRMPO by 95.3 % compared to GBCS. When the configuration size is 1,000 modules, the assignments produced by GBCS and DGA differ a lot. Although the GBCS has the largest global matching gain, its global optimal assignment is non-optimal while its configuration-based search algorithm and link condition algorithm and make it cost more time to find its paths. Compared to GBCS, IMRP reduces transfer consumption by 76.2 % and RSRMPO by 85.3 %, while IMRP also decreases planning time by 90.7 % and RSRMPO by 93.3 %.

Additionally, we extracted all data on reconsfiguration completion rates of GBCS and RSRMPO framework across three configuration sizes of 60, 560, and 1000 modules in Experiment 3 and calculated the average completion rate for each size, as shown in Fig. 10(d). The completion rate of RSRMPO is 1.581 % higher than GBCS in configuartion with 60 modules, and 11.7 % in 1,000 modules. Although the overall costs of the DGA algorithm and the global KM algorithm are approximate, RSRMPO can continue to plan for subsequent modules in the face of existing hollow structures. Simultaneously, we observed that the RSRMPO framework cannot fully ensure the total reconfiguration of

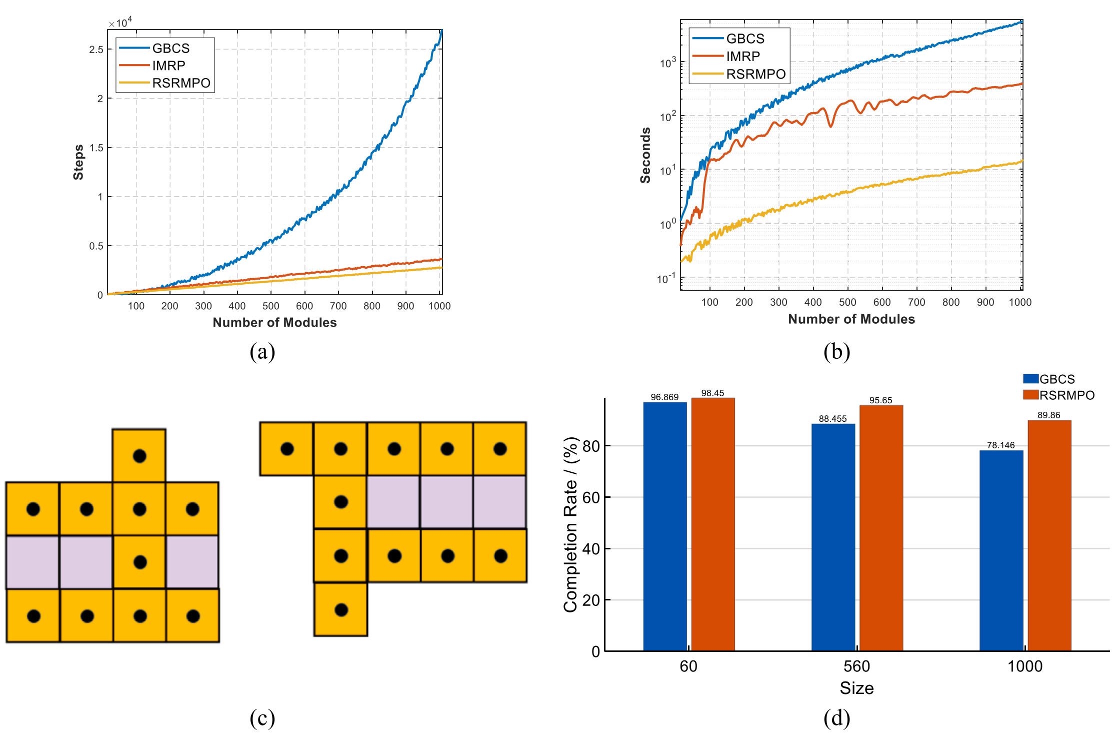


Fig. 10. The experimental results of reconfigurations with random configurations.(a) Schematic cutaway of the 3-D layout, featuring concave structures, with purple grids indicating the target configuration positions. (b) and (e) The first group reveal the effect of the RSRMPO, IMRP and GBCS methods on the performance of reducing SR steps under 50 % initial overlap rate. (c) and (f) The result of RSRMPO, IMRP and GBCS methods in second group. (d) Reconfiguration completion rate of two methods under three different configuration scales.

any target configuration due to the presence of the hollow structure in the original configuration, which causes deadlock of the local SMSRS. Compared to GBCS, IMRP and RSRMPO never formed hollow structure in random concave-structure reconfigurations, which is due to the structure-based priority sequence of the DGA.

Ultimately, we evaluate instances to replicate authentic large-scale SR scenarios in space. Three other experiments demonstrated the SR outcomes of particular configurations: camouflage satellite, space telescope frame, and space manipulator arm on the lunar base, respectively. Fig. 11 illustrates the reconfiguration process of the modular satellite, reconfiguring from the stacked state at launch to the expansive satellite structure with a manipulator, accomplished in 10 reconfiguration cycles. The representative configurations in aerospace engineering were created in BlockBlender utilising 3-D models, as illustrated in Fig. 12.

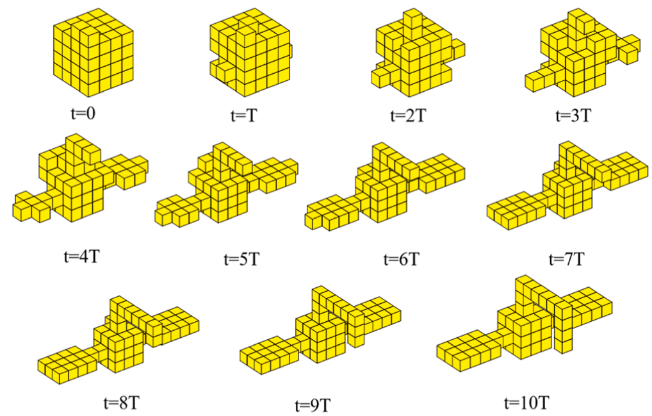


Fig. 11. The SR process by RSRMPO method, which enables multiple modules to undergo parallel self-reconfiguring during the same reconfiguration cycle.

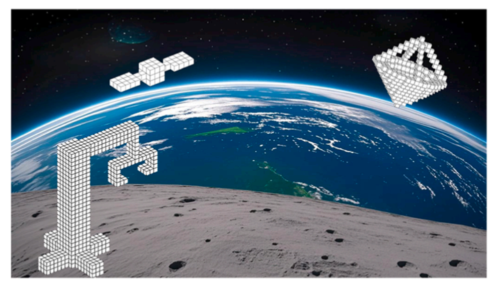


Fig. 12. The representative large-scale SR in aerospace engineering, the camouflage satellite, the space telescope frame, and the space manipulator arm.

Conclusion

In this study, we present a Rapid Self-Reconfiguration Motion Planning Optimization (RSRMPO) method based on the distributed framework to resolve the near-optimal NP-complete challenges associated with preventing the hollow structure formation and increasing efficiency in large-scale reconfiguration for swarm Space Modular Self-Reconfigurable Satellites. In the RSRMPO framework, the virtual connection technique (VCT), which reduces the calculating time of generating movable modules by 51.2 % in 1,000-module random configurations, in conjunction with the Degeneration-Growth Assignment (DGA) method, enables the cube SMSRS pivoting module to swiftly ascertain its connection status and current optimal target position, faster than identifying the graph-link modules by global Depth First Search (DFS) for each reconfiguration. A Feasible Space Map (FS_map) integrated with an energy-cost-optimized A* algorithm only needs to search limited feasible motion space, considering fewer nodes than the pivoting condition and configuration searching calculation in prior works, which reduces path planning time by 67.2 % and takes fewer transfer steps in 1,200-module random configurations compared to the configuration searching method. A priority-based collision avoidance technique can prevent agent deadlock. Simulation experiments confirm that the RSRMPO framework achieves a 93.3 % reduction in planning time, an 85.3 % decrease in energy consumption, and a higher completion rate, 89.85 %, compared to the Graph-Based Configuration Search (GBCS) algorithm in 1,000-module random configurations with 50 % overlap. For 50-module configurations (50 % overlap), it reduced planning time by 95.3 %. These advantages provide near-optimal solutions for reconfiguration scales spanning 10–1,200 modules, demonstrating scalability, computational efficiency, and precision for large-scale space applications. Critically, the RSRMPO framework prevents the formation of hollow structures and unideal target positions during the SR process by the DGA method, but it is unsuitable for existing hollow structures and requires additional improvement. Future research will investigate decentralized strategies that concern local configuration growth trends to minimize reconfiguration transfer steps.

CRediT authorship contribution statement

Lei Chen: Writing – review & editing, Software, Methodology, Formal analysis, Writing – original draft, Supervision, Data curation, Validation, Project administration, Funding acquisition, Visualization, Resources, Investigation, Conceptualization. **Naiming Qi:** Methodology. **Mingying Huo:** Funding acquisition. **Qiufan Yuan:** Resources. **Ze Yu:** Resources. **Wenyu Feng:** Resources.

Declaration of competing interest

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these. There are no conflicts of interest to declare.

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Data availability

The authors do not have permission to share data.

Reference

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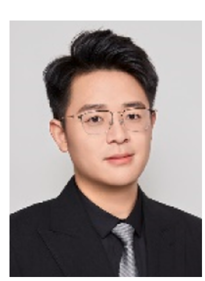

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